

STORYMAPS CHEAT SHEET

What is StoryMaps?

StoryMaps is a drag-and-drop website builder with ArcGIS integration. There are four types of StoryMaps:

Type	Description
Frame	For short form content. Meant to be viewed on phones and mobile devices
Briefing	Presentation style format. Slide-based.
Story	The primary format. Creates a pageless website meant to be viewed on computers, but will adapt to phones and tablets.
Collections	Bundles StoryMaps for sharing.

Building Your StoryMap

Go to <https://storymaps.arcgis.com/content> to begin! Click the ' + Create ' button in the top right corner. The create button will open a menu with options to start a theme, template, and all four types of StoryMaps.

Themes

Sets font, heading styles and primary colors. Themes can be selected in the 'Design' tab of the StoryMap.

Story Basics

Let's look at a **Story**. Everything is added to the Story using content blocks, accessed by clicking the '+' sign on the Story page.

Content blocks are organized under three types

- **Basic:** Text, buttons, separators.
- **Data Visualization:** Link to ArcGIS Online content or auto-generate charts and graphics.
 - Map: Links to maps saved in ArcGIS Online
 - Chart: Input data to generate a chart
 - Infographic: Creates a callout with large text and a icon chosen from presets.
 - Table: Generates a table from inputted data.
- **Media:** Upload images, video, audio and links
 - Unique features:
 - ArcGIS app: Embed other StoryMaps

- **Swipe:** Compares two images with an interactive slider
- **Timeline:** graphic timeline
- **Immersive:** Interactive maps.
 - **Map Tour:** Links ArcGIS Online content overlaid with a text panel. Combines maps, images and text. Best for point data.
 - **Sidecar:** Alternative to Map Tour for ArcGIS Online content, images, video, interactive maps.

Building Content

Planning and Outlining Your Story

Identify Your Target Audience

- What knowledge can you assume they know? What do you assume they don't know?
- How do issues impact your audience?
- Does your Story have a narrative arc your target audience can identify with?

Define Key Takeaways

- 2-4 one-sentence ideas you want your audience to understand, remember, or be able to repeat after reading your Story.

Create a Content Inventory

Your **Content Inventory** includes your written outline, and a visual outline.

- **Written Outline**
 - Consider an opening hook to get your audience interested in your issue.
 - Narrative arcs are a great way to build a throughline when discussing data-heavy issues.
 - Use active and immersive language.
 - Write your copy in a separate document you can copy and paste into the StoryMap.
 - Use placeholders for images and media.
 - Use repetition to drive home your key takeaways.
- **Visual Outline**
 - A list of photos, graphics, charts, images, video and maps
 - Which StoryMap content blocks you will use for your visuals?
 - Use repetition. Similarly structured content blocks group ideas and create throughlines.

Data Visualization

For the "Air Pollution in Milwaukee County" StoryMap, all media was created in R Studio and a vector-based graphics program.

Infographics

- **In Story:** micron size diagram, ozone formation diagram, AQI diagram
- **Programs Used:** Rhino, Inkscape

Charts

- **In Story:** Daily Miles Traveled in Milwaukee County treemap, Reducing Nitrogen Oxide Emissions treemap
- **Programs Used:** R Studio (packages: treemap)

Plotted Maps

- **In Story:** Ozone 70ppb Standard map, Ozone 60ppb Standard map, RSEI Score/Facility map
- **Programs Used:** R Studio (packages: tigris, sf, tmap, tmaptools, cols4all)

Interactive Maps

- **In Story:** Air Quality maps, Health Overlay maps, Heavy Duty VMT map
- **Programs Used:** R Studio (packages: tigris, sf, tmap, tmaptools, cols4all, tidycensus, CDCPLACES, ggmap)

More on Packages

- ggmap was used to register [Stadia Maps](#) basemaps. Stadia needs an api key to be viewable on export
- tmap has a "plot" and "view" mode. View mode will call interactive maps
- tigris pulls geography with geometry. [Walker Data](#) is a great resource for how to utilize tigris
- tidycensus and CDCPLACES are very efficient for wrangling data without web-based dashboards

Tips

Images

- Save all images at 300dpi.
- Inline images in StoryMaps have limited customization. Making quick placeholder images to test dimensions was the best way to avoid headaches with this.
- Larger images should be >1920px in one direction (usually horizontal).

Phone Friendly

- Sidecar allows you to upload static images of interactive content, which appear broken on phones. Find

these options by clicking the gear icon once you've added media to a Sidecar.

- Click 'Include a button in the embed page to open live content in a new tab', also accessible in the Sidecar options, to allow interactive Sidecar media to be viewed on phones.
- The 'Preview' option in the StoryMaps header lets you view your Story on different sized screens.

Accessibility

- Use high contrast between background colors and text colors.
- Check contrast of media by viewing it in black and white.
- Use sans serif font.
- Add alt text to all media. Adding alt text is accessible in the options of your content blocks. The 'Insights' tab in the header of your Story will also flag accessibility issues.
- In R Studio, the cols4all package has a dashboard for selecting map bin colors, call `c4a_gui()` to view the dashboard. Palettes are tagged for legibility and colorblind friendliness.

Limitations

- Limited text formatting options
- Limited inline image customization
- Difficult to share or collaborate on StoryMaps without an ArcGIS organizational subscription
- Sidecar feature can be difficult to navigate for end users.

Resources

[The R Graph Gallery](#)

[Hosting html maps on GitHub](#)

[Leaflet basemap samples](#)

[tmap Documentation](#)

[Thematic Maps in tmap](#)